SANJIVANI UNIVERSITY

School of Engineering and Technology

Department of Artificial Intelligence & Data Science

UNIT – 1 MINI PROJECT REOPRT ON

"Enterprise Student Record and Academic Management Platform"

Design Thinking and Product Planning

Submitted by:

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CERTIFICATE

This is to certify that Ms. Deokar Gayatri Sandip, PRN 2126UDSF2536, student of DSY 2nd Year, Department of Artificial Intelligence & Data Science, Sanjivani University, Kopargaon, has successfully completed the Unit–1 Mini Project titled "Enterprise Student Record and Academic Management Platform – Design Thinking and Product Planning" under the guidance of Karan Gaikwad Sir (IBM).

The work presented in this report follows the Unit–1 mini-project activities and deliverables prescribed for business understanding, design thinking, IBM Design Thinking, stakeholder analysis, user personas, requirements engineering, product vision, MVP definition, roadmap, project planning, and initial system architecture.

Role	Name	Signature
Project Guide	Karan Gaikwad Sir (IBM)	_____
Head of Department	K. P. Jadhav Sir	_____
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Date: _____ Place: Kopargaon

DECLARATION

I Deokar Gayatri Sandip, PRN 2126UDSF2536, student of DSY 2nd Year – Artificial Intelligence & Data Science at Sanjivani University, Kopargaon, hereby declare that this Unit–1 Mini Project Report titled "Enterprise Student Record and Academic Management Platform – Design Thinking and Product Planning" has been prepared by me for academic submission.

The report documents the product-planning work expected in Unit–1. The project begins with understanding the academic-management domain rather than directly starting implementation. It identifies users and stakeholders, studies their needs and pain points, defines functional and non-functional requirements, applies Design Thinking and IBM Design Thinking, defines an MVP, prepares a product roadmap, and proposes an initial system architecture.

I have used the supplied Unit–1 mini-project guide and training/project reference document as the basis for the structure and terminology of this report.

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I am thankful to my project guide, Prof. Karan Gaikwad Sir (IBM), for providing guidance and direction regarding Design Thinking, product planning, requirements engineering, stakeholder identification, and the overall software product development lifecycle.

I also express my gratitude to K. P. Jadhav Sir, Head of Department, and Dean Kavitha Rani P. for their academic support and encouragement.

Finally, I thank my teachers, classmates, and everyone who contributed directly or indirectly to the understanding and planning of this project.

Deokar Gayatri Sandip

ABSTRACT

The Enterprise Student Record and Academic Management Platform is a proposed software product for managing student records and academic information within a higher education institution. The project is developed as a Unit–1 mini project with emphasis on product discovery, business understanding, Design Thinking, IBM Design Thinking, stakeholder analysis, requirements engineering, MVP planning, and product roadmap creation.

The central idea is to reduce the difficulties associated with fragmented or manual academic record management. The proposed platform brings student registration, academic records, attendance, examination information, results, and report generation into a structured digital system. The project deliberately focuses on understanding the problem and planning the product before implementation.

The report follows the five Design Thinking stages: Empathize, Define, Ideate, Prototype, and Test. It also documents IBM Design Thinking principles of Focus on User Outcomes, Restless Reinvention, and Diverse Empowered Teams, along with the Observe → Reflect → Make loop.

The final planning output includes a Business Requirements Document, product vision, stakeholder matrix, personas, functional and non-functional requirements, MVP scope, roadmap, initial architecture, project charter, repository structure, and product-discovery documentation.

Keywords: Student Records, Academic Management, Design Thinking, IBM Design Thinking, Requirements Engineering, MVP, Product Roadmap, Software Engineering.

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1. Introduction and Objective

The Enterprise Student Record and Academic Management Platform is a proposed software product for managing student records and academic information within a higher-education institution. The Unit-1 brief requires students to understand the educational domain before coding, identify stakeholders, analyze user needs, define software requirements, and prepare a product vision and roadmap.

1.1 Objective

- Understand the academic management process.
- Study student admissions, course registration, attendance, examination and result management.
- Identify problems associated with manual record management.
- Apply the five Design Thinking stages: Empathize, Define, Ideate, Prototype and Test.
- Apply IBM Design Thinking principles and document the Observe → Reflect → Make cycle.
- Identify stakeholders and develop user personas.
- Define functional and non-functional requirements.
- Prepare product vision, MVP, roadmap, architecture concept and project plan.

1.2 Scope of this Report

This report is a product-planning and Design Thinking document. It does not claim that a complete production application has already been implemented. The architecture, MVP and repository sections describe the planned product structure for subsequent Agile planning and development.

2. Business Understanding

The platform is intended for the academic record-management process of a higher-education institution. The business workflow considered in this mini project covers the activities explicitly identified in the assignment: student admissions, course registration, attendance, examination and result management.

2.1 Core Academic Operations

Operation	Purpose	Typical Information Managed
Student Admission / Registration	Create and maintain student academic identity.	Student profile, program, semester, registration details
Course Registration	Record the courses/subjects associated with a student.	Course, semester, faculty, enrollment
Attendance Management	Record and review attendance information.	Attendance entries, percentage, subject-wise status
Examination Management	Manage examination-related academic information.	Exam, subject, marks/grades, assessment records
Result / Academic Report	Generate academic information and reports.	Semester results, grades/marks, academic summaries
Administration	Maintain records and support academic workflows.	User access, records, reports, audit information

2.2 Manual Record Management Challenges

- Repeated manual entry and maintenance of student information.
- Difficulty in maintaining consistent academic records across processes.
- Time-consuming retrieval of student records.
- Greater possibility of record-entry and reporting errors.
- Difficulty in generating academic reports for large numbers of students.
- Limited centralized visibility for academic administrators.

2.3 Proposed Business Response

The proposed platform provides a centralized approach to student records and academic information. It is planned around role-based users, structured records, attendance, examination/results and reporting so that the institution can move from fragmented/manual handling toward an organized digital workflow.

3. Problem Statement and Business Requirements

3.1 Core Problem Statement

Students, faculty members and academic administrators need a centralized and organized way to manage student records and academic information because fragmented or manual record handling can make registration, attendance, examination, result management and report generation time-consuming and error-prone.

3.2 Business Requirements

ID	Business Requirement	Expected Business Need
BR-01	Centralized student records	Maintain student academic information in an organized platform.
BR-02	Registration management	Support student and course/academic registration information.
BR-03	Attendance management	Maintain attendance information for academic monitoring.
BR-04	Examination and results	Maintain examination and result information.
BR-05	Academic reporting	Generate academic reports from maintained records.
BR-06	Role-based access	Allow appropriate users to access relevant functions.
BR-07	Reliable records	Improve consistency and reliability of stored information.
BR-08	Future scalability	Allow the planned product to evolve through Agile development.

3.3 Business Outcome

The intended outcome is a well-planned software product that can move into Agile planning and later development. The Unit-1 objective emphasizes planning the product before implementation.

4. Stakeholder Analysis

The Unit-1 brief identifies students, faculty members, academic administrators, examination cell staff and IT support personnel as key stakeholders.

Stakeholder	Role / Interest	Expected Need
Students	Use academic services and view their records	Easy access to relevant academic information.
Faculty Members	Maintain academic information and attendance	Efficient record entry and retrieval.
Academic Administrators	Manage academic processes and records	Centralized visibility and reporting.
Examination Cell Staff	Handle examination/result information	Organized examination and result records.
IT Support Personnel	Support the software environment	Maintainability, reliability and controlled access.

4.1 Stakeholder Responsibility Matrix

Stakeholder Group	Responsibilities / Expectations
Business stakeholders	Define academic-management needs and validate usability.
Academic users	Use and maintain relevant academic records.
Engineering team	Design and develop the planned system.
IT support	Support deployment, maintenance and system availability.

4.2 Stakeholder Priority for Discovery

Discovery should begin with users who directly experience academic-record workflows. Their needs should be converted into requirements and validated through prototypes before detailed implementation.

5. User Persona Development

Personas represent the main user categories identified in the Unit-1 brief. They help the team keep product planning focused on user needs.

Persona	Goals	Responsibilities	Challenges / Expectations
Student	Access academic information and records efficiently.	Use academic services and review relevant information.	Wants clear, accessible and organized information.
Faculty Member	Maintain academic information efficiently.	Maintain relevant academic records and attendance.	Wants simple data entry and retrieval.
Academic Administrator	Manage academic information and generate reports.	Coordinate academic records and processes.	Needs centralized visibility and reporting.
Examination Cell Staff	Manage examination and result information.	Maintain examination/result records.	Needs organized records and accurate report generation.
IT Support Personnel	Keep the platform reliable and maintainable.	Support system operation and technical issues.	Needs security, reliability and maintainability.

5.1 Persona-Based Product Focus

- Student: accessibility and understandable academic information.
- Faculty: efficient academic-record management.
- Administrator: centralized information and reporting.
- Examination Cell: organized examination and result workflows.
- IT Support: security, reliability and maintainability.

6. Design Thinking Report

Design Thinking is applied as an iterative, human-centered problem-solving approach. The five stages used in this assignment are Empathize, Define, Ideate, Prototype and Test.

Stage 1 : Empathize

Understand the users and their academic workflows. Focus on students, faculty members and academic administrators. Observe how they access, enter, verify and report academic information.

6.1 Observe Academic Workflows

Observation Area	Questions to Understand
Registration	What information is entered? Where is it maintained?
Attendance	How is attendance recorded and retrieved?
Examination	How is examination information maintained?
Results	How are results recorded and reported?
Reports	Which academic reports are required and who uses them?
Support	What difficulties occur while accessing or maintaining records?

6.2 User Pain-Point Themes

- Repeated or fragmented record handling.
- Difficulty locating information quickly.
- Manual effort in report preparation.
- Potential inconsistency or data-entry errors.
- Need for role-appropriate access.

7. Design Thinking – Stage 2: Define

Convert observed needs into a clear problem definition.

- Core problem: academic information is difficult to manage efficiently when records workflows are disconnected.
- Need: a centralized platform with role-specific academic functions.
- Outcome: users should be able to complete common academic-record tasks with less repetitive effort.

7.1 Defined Problem

Academic institutions need a centralized student-record and academic-management platform that organizes registration, attendance, examination, result and reporting information while providing appropriate access to students, faculty, administrators, examination staff and IT support.

7.2 User Need Statements

User	Need Statement
Student	Needs clear and organized access to relevant academic records.
Faculty	Needs an efficient way to maintain academic information.
Administrator	Needs centralized records and academic reports.
Examination Cell	Needs organized examination and result information.
IT Support	Needs a secure, reliable and maintainable system.

7.3 Design Goal

Design a product around user outcomes and academic workflows rather than beginning with code or technology selection.

8. Design Thinking – Stage 3: Ideate

The Ideate stage generates solution possibilities before narrowing the scope to an MVP.

Idea	Purpose
Centralized student profile	Maintain core student information in one place.
Academic record dashboard	Provide organized access to academic information.
Course/subject management	Maintain course and subject registration information.
Attendance module	Record and retrieve attendance information.
Examination and result module	Maintain examination and result records.
Academic report generation	Generate reports from stored academic information.
Role-based access	Provide relevant access for different user groups.
Search and filtering	Make student records easier to locate.
Future notification capability	Support future academic communication workflows.

8.1 Idea Selection Principle

For the first release, features should be selected according to user need, business value and feasibility. Non-essential features can be deferred so that the team can validate the core product first.

9. Design Thinking – Stage 4: Prototype

The Prototype stage represents the proposed solution using low-fidelity wireframes or screen concepts before full implementation.

9.1 Proposed Core Screens

Screen	Main Content
Login	User authentication and role-based entry.
Student Dashboard	Student profile and academic information.
Faculty Dashboard	Academic-record and attendance functions.
Administrator Dashboard	Student records, management and reports.
Examination/Results	Examination and result information.
Reports	Academic report generation and viewing.

9.2 Low-Fidelity Prototype Flow

LOGIN → ROLE IDENTIFICATION → DASHBOARD → SELECT ACADEMIC MODULE → VIEW / ENTER RECORD → SAVE → REPORT / VIEW RESULT

9.3 Prototype Principle

The prototype should be simple enough to change after feedback. It is intended to validate navigation, information organization and task flow before expensive implementation.

10. Design Thinking – Stage 5: Test

- The Test stage validates the prototype with representative users. Testing should focus on whether users can understand the navigation and complete intended academic-record tasks.
- Show the prototype/workflow to representative users and collect feedback on clarity, navigation and task completion.

- Check whether users understand the dashboard.
- Check whether common tasks are easy to locate.
- Identify confusing fields or unnecessary steps.
- Refine the prototype before detailed development.

Test Area	**Validation Question**
Login	Can the user understand how to enter the platform?
Navigation	Can the user locate the required academic module?
Record Entry	Can the intended information be entered clearly?
Record Retrieval	Can a student/academic record be found?
Reports	Can the user understand and generate the required report?
Usability	Is the interface understandable for the target user?

10.1 Feedback Loop

Test feedback should be documented, reflected upon and used to improve the prototype. This creates an iterative process rather than treating testing as a final one-time activity.

10.2 Design Thinking Summary

Empathize → understand users; Define → frame the problem; Ideate → generate solutions; Prototype → represent the solution; Test → validate and improve.

11. IBM Design Thinking

The Unit-1 brief requires application of three IBM Design Thinking principles: Focus on User Outcomes, Restless Reinvention, and Diverse Empowered Teams.

11.1 Focus on User Outcomes

The product should focus on what users need to accomplish. For example, instead of only defining a technical goal such as creating a student-record database, the team should focus on enabling users to access and manage required academic information effectively.

11.2 Restless Reinvention

The product should be treated as continuously improvable. Feedback, usage observations and changing academic needs can lead to refinement of workflows, interfaces and product priorities.

11.3 Diverse Empowered Teams

Product planning benefits from collaboration among domain users, product planners, UI/UX contributors, software developers, database contributors, testers and IT/DevOps support.

11.4 Application to This Project

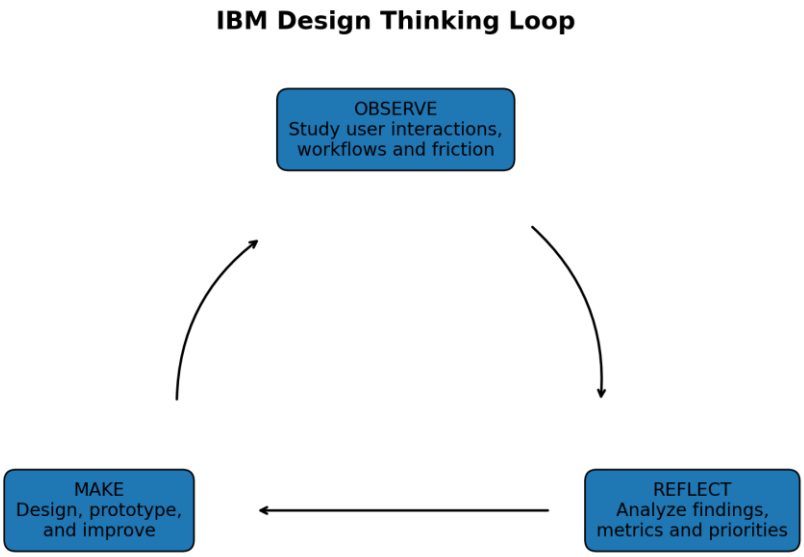
IBM Principle	Application
Focus on User Outcomes	Measure success through useful academic workflows and user needs.
Restless Reinvention	Use feedback to refine the product iteratively.
Diverse Empowered Teams	Bring academic and technical perspectives into product decisions.

❖ User Outcome Examples

- Student outcome: access relevant academic information through a clear and understandable interface.
- Faculty outcome: maintain attendance and academic records with less repetitive effort.
- Administrator outcome: retrieve records and generate reports efficiently.
- IT support outcome: maintain a structured and supportable system.

12. IBM Loop Documentation

The IBM Loop is documented as Observe → Reflect → Make. The loop supports continuous product discovery and improvement.



12.1 Observe

- Observe academic workflows and user interactions.
- Identify friction in registration, attendance, examination, results and reporting.
- Collect feedback from the relevant stakeholder groups.

12.2 Reflect

- Analyze observations and user feedback.
- Identify which problems are most important for the current product scope.
- Align product priorities with user outcomes and business needs.

12.3 Make

- Create or modify a prototype.
- Convert validated needs into product requirements.
- Prepare the next development or design iteration.

12.4 Continuous Cycle

The loop is not a linear end point. After making a change, the team observes its effect and begins the cycle again.

13. Functional Requirements Specification (FRS)

The Unit-1 brief asks for functional requirements such as student registration, attendance management, result generation and report generation.

ID	Functional Requirement
FR-01	User authentication and role-based access for relevant users.
FR-02	Student registration and student record management.
FR-03	Course/subject information management.
FR-04	Attendance management.
FR-05	Examination information management.
FR-06	Result management and result generation.
FR-07	Academic report generation.
FR-08	Search and retrieval of student academic information.
FR-09	Administrative management of relevant academic records.
FR-10	Controlled access to functions according to user role.

13.1 Functional Flow

Authenticate User → Identify Role → Open Role Dashboard → Select Academic Function → View/Add/Update Information → Validate → Save → Generate/View Required Output

13.2 FRS Validation

Each functional requirement should later be converted into user stories, acceptance criteria and test cases during Agile planning.

14. Non-Functional Requirements Specification (NFRS)

The Unit-1 brief identifies security, usability, reliability and scalability as important non-functional requirements.

ID	Category	Requirement
NFR-01	Security	Protect academic information through appropriate authentication and access control.
NFR-02	Usability	Provide clear navigation and understandable workflows for intended users.
NFR-03	Reliability	Maintain consistent and dependable academic records.
NFR-04	Scalability	Allow the product to support growth in users and academic records.
NFR-05	Maintainability	Keep the product structure organized for future development and maintenance.
NFR-06	Data Integrity	Prevent inconsistent or invalid academic records through validation.
NFR-07	Availability	Make academic information accessible when required by authorized users.

14.1 Why NFRs Matter

A system can provide required functions but still fail to meet user expectations if it is insecure, difficult to use, unreliable or difficult to scale. Therefore NFRs are part of product planning, not an afterthought.

15. Product Vision

15.1 Vision Statement

To provide a centralized and user-focused Student Record and Academic Management Platform that organizes academic information, simplifies academic-record workflows, and supports efficient access to student registration, attendance, examination, results and reporting.

15.2 Mission Statement

To understand academic users, design around their needs, validate the product iteratively, and prepare a maintainable software product that can progress from product discovery to Agile development.

15.3 Product Objectives

- Centralize student and academic information.
- Reduce unnecessary manual effort in academic-record workflows.
- Support organized attendance, examination and result management.
- Enable academic reporting from maintained records.
- Provide appropriate access to different stakeholder groups.
- Create a product foundation suitable for future Agile development.

15.4 Success Criteria

Success should be evaluated through user feedback, usability of the core workflows, completeness of required records, reliability of information and successful completion of planned MVP functions.

16. Minimum Viable Product (MVP)

The Unit-1 brief asks the team to define the first release. The MVP should contain the core capabilities needed to validate the product concept.

MVP Feature	Description
Student Registration	Create and maintain student registration information.
Student Record Management	Maintain relevant academic student records.
Attendance Management	Record and retrieve attendance information.
Examination Management	Maintain examination-related information.
Result Management	Maintain and generate result information.
Academic Report Generation	Generate core academic reports.
Role-Based Access	Provide relevant access according to user role.

16.1 Deferred / Future Features

The source material does not prescribe specific deferred features for this student platform. Therefore, additional capabilities should be selected after stakeholder discovery and MVP validation rather than being treated as confirmed requirements.

16.2 MVP Principle

The MVP should be small enough to build and test while still representing the central value of the product.

17. Product Roadmap

The product roadmap is aligned with the course progression and the Unit-1 requirement to prepare a phased roadmap.

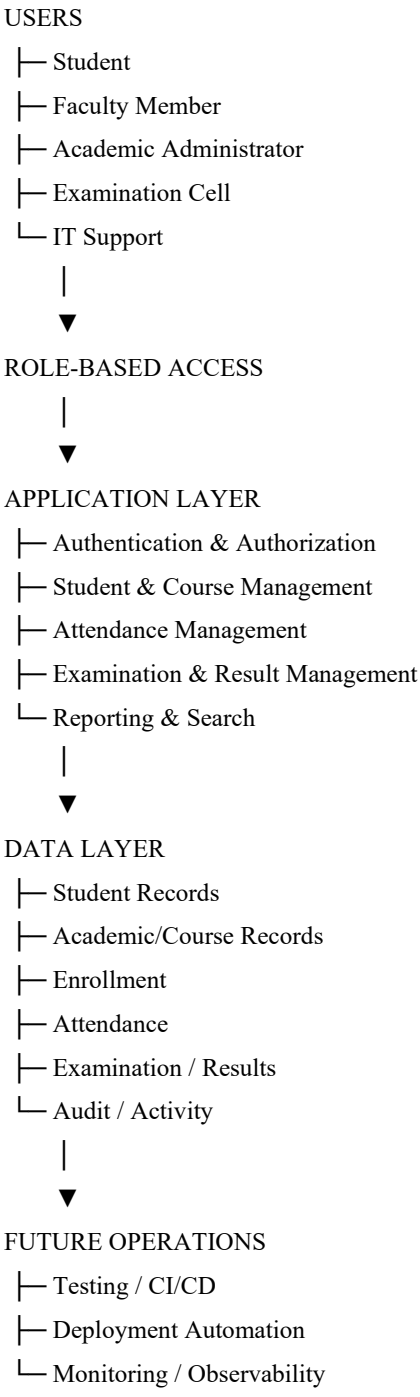
Phase	Focus	Expected Output
Phase 1	Domain Analysis & Problem Understanding	Business understanding, stakeholders and problem definition.
Phase 2	Design Thinking & Prototyping	Personas, ideas, prototype and testing plan.
Phase 3	Requirements & Agile Planning	FRS, NFRS, backlog-ready requirements and priorities.
Phase 4	Core Application Development	Student records, attendance, examination/results and reporting.
Phase 5	Testing & Quality Improvement	Validation of functional and non-functional requirements.
Phase 6	Deployment / Operations Planning	Preparation for operational use and technical support.
Phase 7	Continuous Improvement	Observe, reflect and make based on feedback.

17.1 Roadmap Principle

The roadmap is phased so that discovery and design occur before implementation, followed by iterative development and improvement.

18. Initial System Architecture

The Unit-1 deliverables include an initial system architecture diagram. At this stage it is a conceptual architecture rather than a final implementation design.



18.1 Architecture Responsibilities

Layer	Responsibility
Users	Interact with the academic-management product.
Presentation	Provide role-appropriate screens and workflows.
Application	Process academic-management operations.
Data	Store structured student and academic information.
Supporting Services	Address security, validation, reliability and operational needs.

18.2 Architecture Principles

- Separate responsibilities into modules.
- Protect academic information through role-based access.
- Keep the architecture extensible.

Plan DevOps automation after core requirements are understood.

19. Project Charter

Charter Element	Planned Definition
Project Title	Enterprise Student Record and Academic Management Platform
Purpose	Plan a software product for student records and academic information management.
Primary Users	Students, faculty members, academic administrators, examination cell staff and IT support.
Scope	Business understanding, Design Thinking, requirements, product planning and architecture concept.
MVP	Registration, student records, attendance, examination/results and academic reporting.
Major Deliverables	BRD, Product Vision, Stakeholder Analysis, Personas, FRS, NFRS, Design Thinking, IBM documentation, MVP, roadmap, architecture, repository and discovery documentation.
Constraints	Unit-1 is focused on planning before coding.
Success	A coherent product plan ready for Agile planning and later development.

19.1 Milestone Plan

Milestone	Output
M1	Business understanding and problem statement
M2	Stakeholder analysis and personas
M3	Design Thinking documentation
M4	IBM Design Thinking and Loop
M5	FRS and NFRS
M6	Vision and MVP
M7	Roadmap and architecture
M8	Project charter and repository structure
M9	Final product discovery documentation

20. Project Repository Initialization and Folder Structure

20.1. Repository Structure

The training reference provides a standardized repository structure containing application code, tests, documentation, deployment configuration, infrastructure automation, and monitoring configuration. □filecite□turn0file0□L349-L365□

```
student-academic-management/
├── .github/ # CI/CD workflows
├── app/ │
├── modules/ # Students, Attendance, Exams, Results │
├── database/ # Schemas and models
│ └── tests/ # Unit and integration tests
├── docs/ # BRD, FRS, NFRS, design documents
├── docker/ # Container configuration
├── deployment/ # Deployment manifests
├── monitoring/ # Monitoring configuration
└── README.md # Project setup and usage
```

21. Product Discovery Documentation

Product discovery records the reasoning that connects the business problem to the proposed product. The following summary can be used as the Unit-1 discovery record.

Discovery Area	Finding / Decision
Target Domain	Higher-education student academic record management
Primary User Groups	Students, faculty, academic administrators; examination and IT staff are supporting stakeholders
Core Problem	Academic information and workflows can be difficult to manage efficiently when records/processes are disconnected or manual
User Needs	Fast access, accurate information, clear workflows, controlled access and useful reports
Core Solution	Centralized platform for registration, records, attendance, examinations/results and reports
Design Approach	Design Thinking + IBM Design Thinking + Observe → Reflect → Make
MVP Decision	Start with core academic record functions and role-based access
Future Direction	Iterate through user feedback and add integrations, notifications, analytics and deployment capabilities as appropriate

22. Conclusion

The Enterprise Student Record and Academic Management Platform has been planned as a user-focused software product for managing student records and academic information in a higher-education environment.

The report covers the activities and deliverables: business understanding, Design Thinking, IBM Design Thinking, IBM Loop, stakeholder analysis, user personas, functional and non-functional requirements, product vision, MVP, roadmap, initial system architecture, project charter, repository structure and product discovery.

22.1 Final Deliverable Checklist

Deliverable	Included in Report
Business Requirements Document (BRD)	Yes
Product Vision Document	Yes
Stakeholder Analysis Report	Yes
User Persona Report	Yes
Functional Requirements Specification (FRS)	Yes
Non-Functional Requirements Specification (NFRS)	Yes
Design Thinking Report	Yes
IBM Design Thinking Report	Yes
IBM Loop Documentation	Yes
MVP Definition	Yes
Product Roadmap	Yes
Initial System Architecture Diagram	Yes
Project Charter	Yes
Project Repository Initialization Plan	Yes
Project Folder Structure	Yes
Product Discovery Documentation	Yes